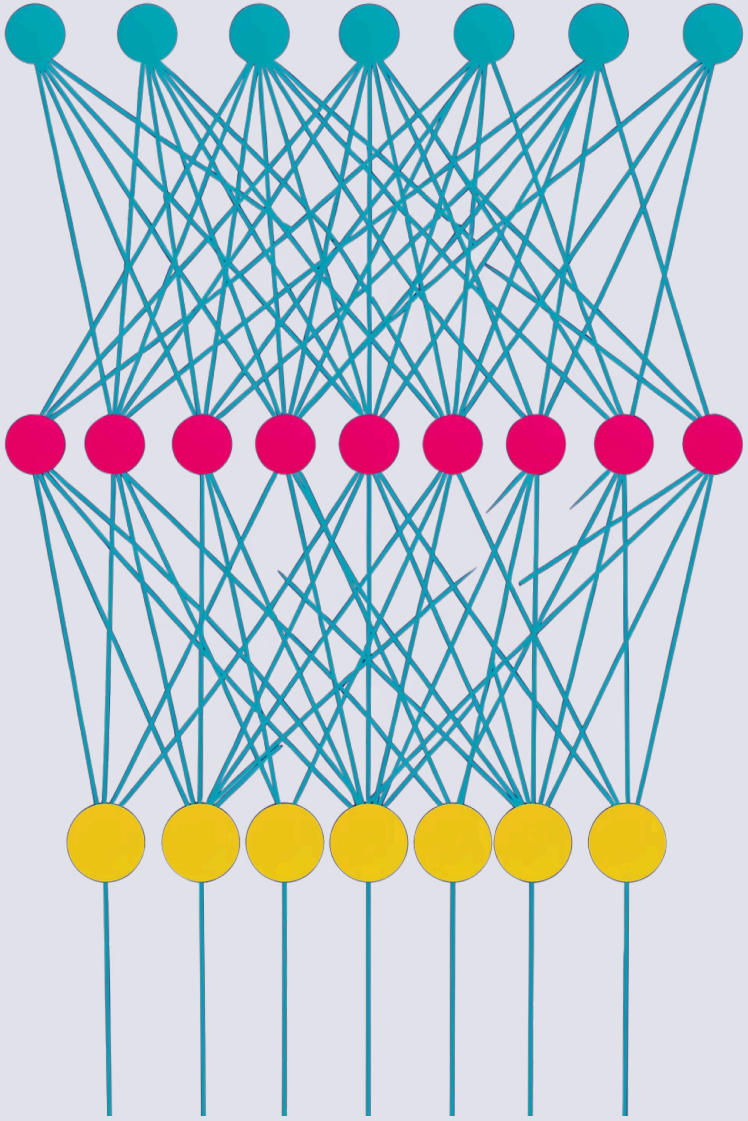


Why Transformers Need Position Numbers

Order, speed, and the trade the Transformer made

Introduction to Generative AI

The Core Idea, Isolated

Two ways to read a sentence:

Old approach (RNNs):	**Transformer approach:**
one word at a time	all words at the same time

Old Approach: RNNs

Read one word, then the next, then the next, **in order**

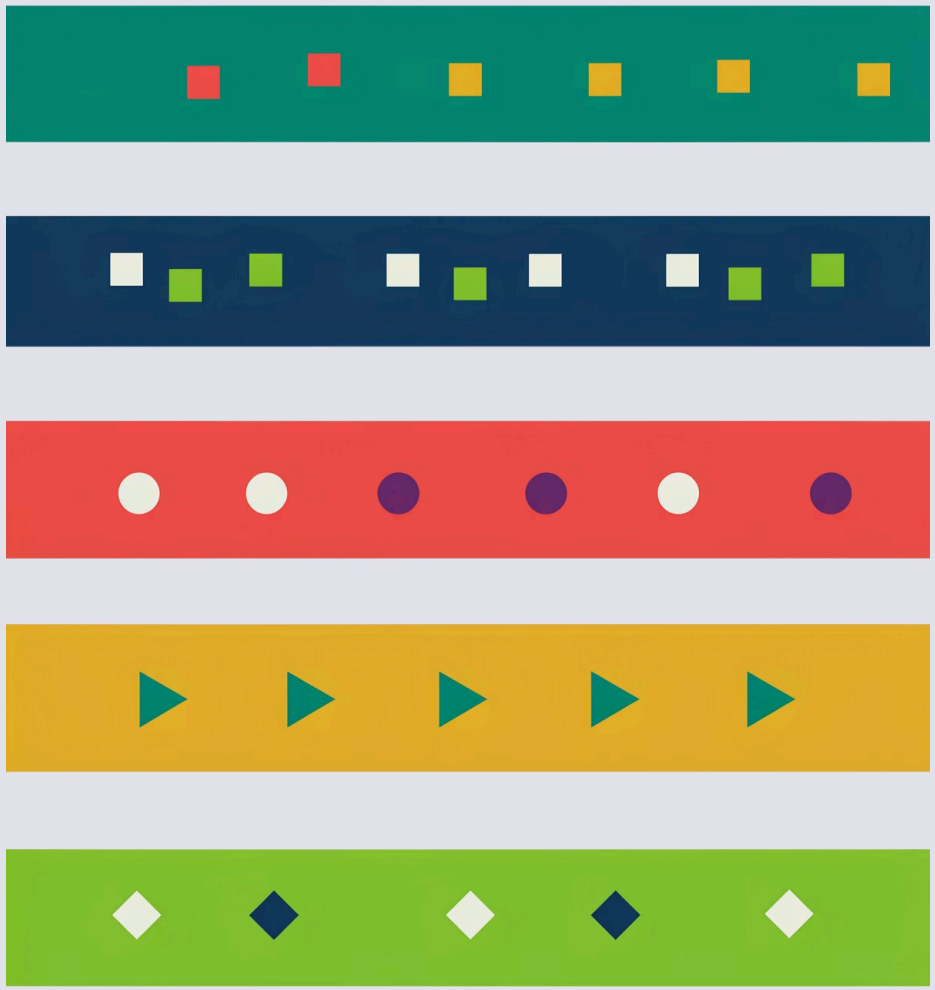
Like reading a book left to right

Word 1 is processed before word 2, before word 3

So the model knows the order just by **when** it saw each word

No extra information needed.





Transformer Approach

Dump all the words in **at the same time**, as one batch

Nothing is processed "first" or "second"

Everything is looked at **simultaneously**

This is **faster**, because words are processed in parallel. But it creates a problem.

The Problem It Creates

With no sense of order, these two sentences look **identical** to the model:

"The dog bit the man"

"The man bit the dog"

Same words. **Completely different meaning.**



Why That Is a Disaster

If the model only sees a pile of words:

> the, dog, bit, the, man

it has no way to tell the two sentences apart.

That is obviously a disaster for understanding language.

The Fix: Position Numbers

The model tags each word with a label saying where it sits:

01	02	03
"I'm word #1"	"I'm word #2"	...and so on

Even though all the words arrive together, the model can still tell the order.

Position Numbers in Action

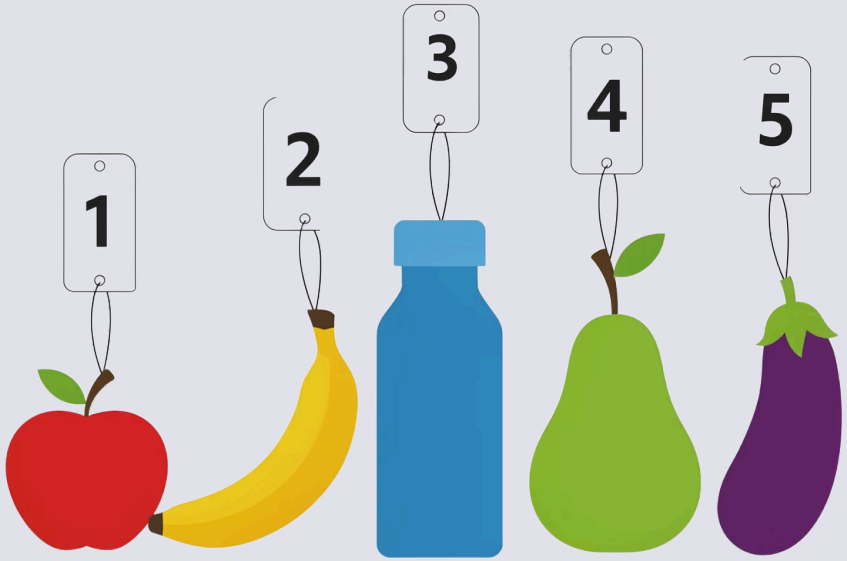
"dog" (position 2) bit "man" (position 5)

"man" (position 2) bit "dog" (position 5)

Same words, different positions, **different meaning**.

The One-Line Version

- Reading one word at a time gives you order **for free**
- Reading all words at once is **faster**, but throws the order away
- So the model must be **told** the order separately, using position numbers



What Position Numbers Are (and Are Not)

They are **not** there to help the model understand meaning

They are purely there to **restore ordering information**

That information was lost the moment the model started reading in parallel

So Why Not Just Read Word by Word?

Two reasons:

1. Speed

(training)

2. Forgetting

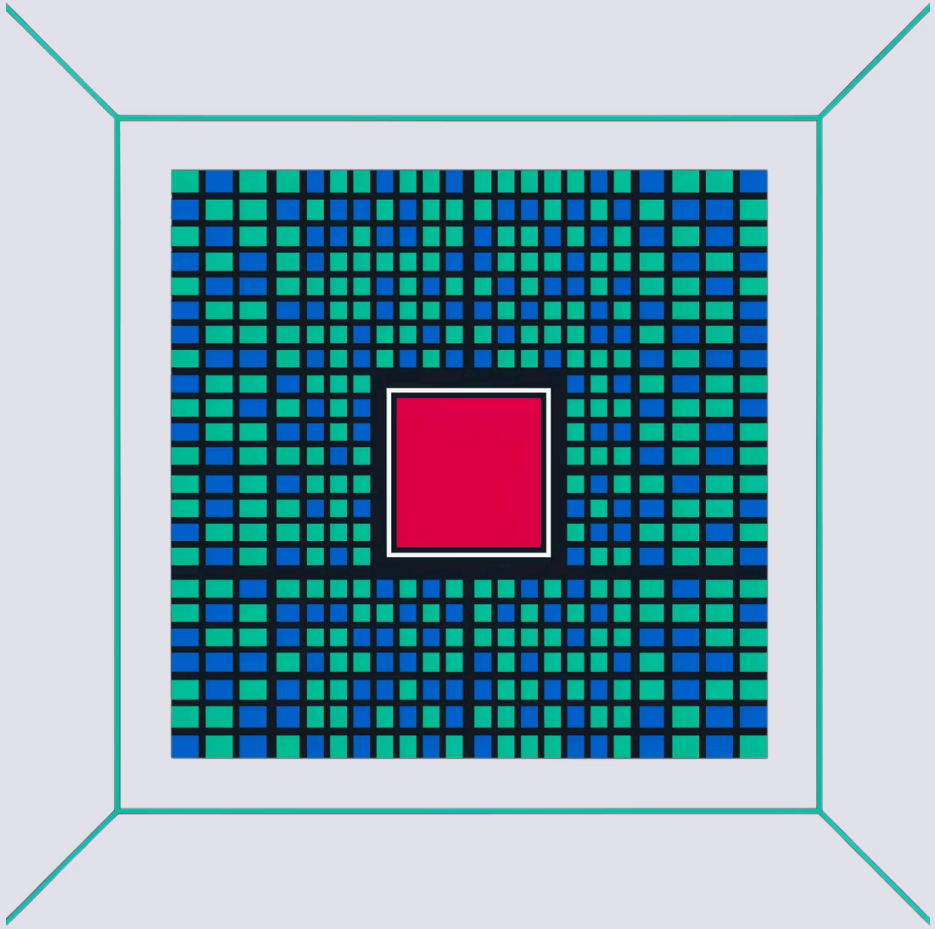
(the bigger one)

Reason 1: Speed

Word-by-word is a **chain**:

1. **Word 5 waits for word 4**
2. **Word 4 waits for word 3**
3. **...and so on**

Each step waits for the previous one.



Why the Chain Is a Problem

Training means processing **millions of sentences**

GPUs are built to do **thousands of things simultaneously**

A strict sequence cannot use that power

Reading all words at once lets the model process a whole sentence **in parallel**

Result: dramatically faster to train.



Reason 2: Forgetting

Imagine reading a 40-word sentence one word at a time.

After each word, you must carry everything understood so far into a **single running summary**, then discard the actual words.

A Summary of a Summary of a Summary...



By word 40, the model relies on a summary of a summary... **39 times over**



Details from the beginning get **diluted or lost**



This is a real, well-documented weakness of RNNs



Fine for short sentences; on long ones they start to **forget** the early parts

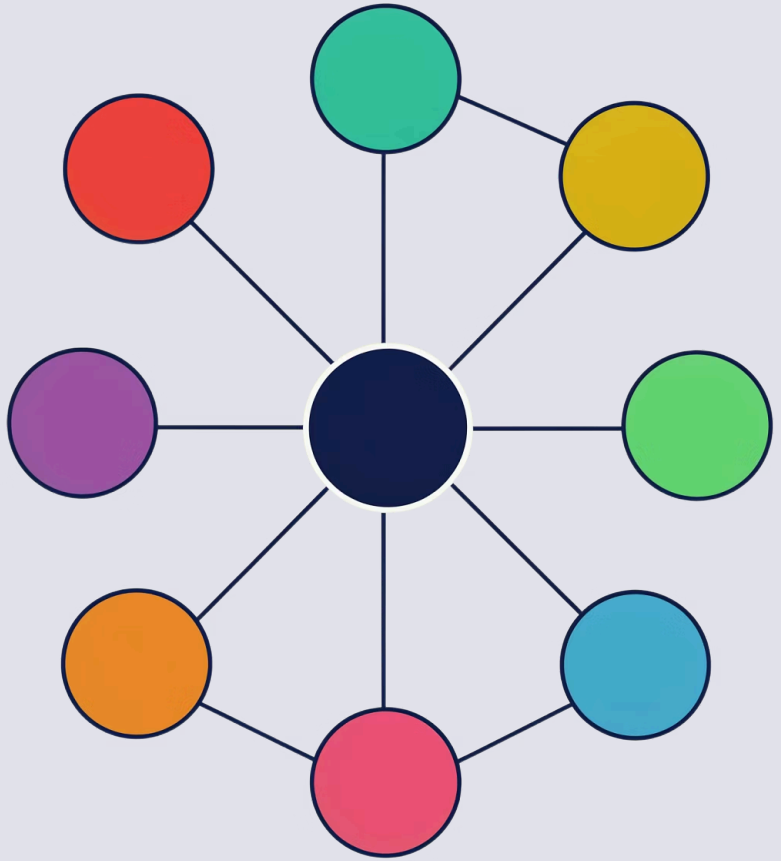
What the Transformer Does Instead

Because it reads everything at once, **any word can look directly at any other word**

No matter how far apart they are

No passing information through 39 intermediate steps

Word 40 can look at word 1 as easily as at word 39. Nothing gets diluted.



The Tradeoff, Stated Plainly

	Word-by-word (RNN)	All-at-once (Transformer)
Order	Free (comes from reading sequence)	Lost; must be re-added with position numbers
Speed	Slow; must wait for each word	Fast; all words processed in parallel
Long sentences	Weak; early words get "forgotten"	Strong; every word connects directly to every other word

A Small Price to Pay

The Transformer gives up **"order for free"**. In exchange it gains:

Speed

A model that does not forget the start of a long sentence

That is the trade its designers made, and **it is why it won out**.

Key Takeaways

1

RNNs read word by word, so order comes **for free**

2

Transformers read all words at once, so order is **lost**

3

Position numbers restore the order

4

The trade is worth it: **faster training** and **no forgetting** over long distances



Quick Check: Discuss in Pairs

1. Why would "The dog bit the man" and "The man bit the dog" look identical without position numbers?

2. Do position numbers help the model understand meaning? Why or why not?

3. Why can't an RNN take full advantage of a GPU?

4. In a 40-word sentence, how does the Transformer connect word 40 to word 1?